02443 Stochastic Simulation

Day 2: Sampling from discrete distributions

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Agenda

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Aim

Day 1

We learned about random number generation and statistical tests to evaluate the generated numbers. Therefore, we assume that we have access to a supply of independent sample from a continuous uniform distribution on $(0, 1)$.

Day 2

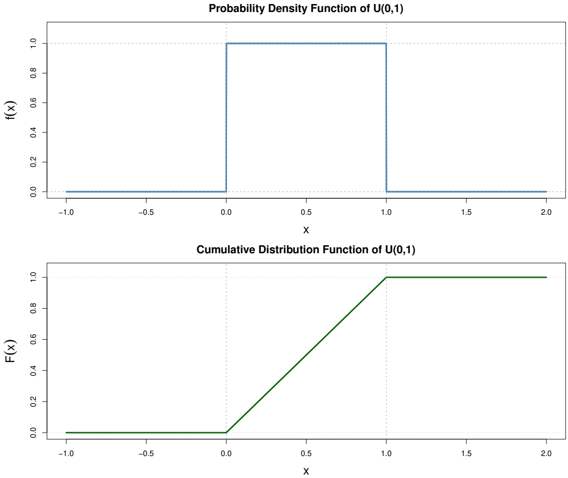
We learn about methods to transform the random numbers into random variables from various discrete and continuous distributions.

Goal: To generate random variables with a given distribution.

The uniform distribution

The uniform distribution will serve as the building block.

Recap



PDF:
$$f(x) = 1, \quad 0 \leq x \leq 1$$

CDF:
$$F(x) = x, \quad 0 \leq x \leq 1$$

Expectation:
$$\mathbb{E}[U] = \frac{1}{2}$$

Variance:
$$\mathbb{V}[U] = \frac{1}{12}$$

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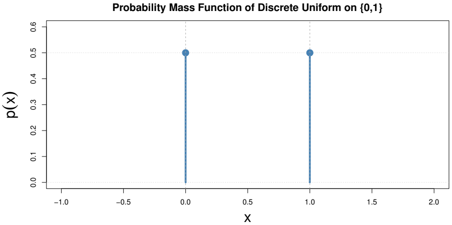
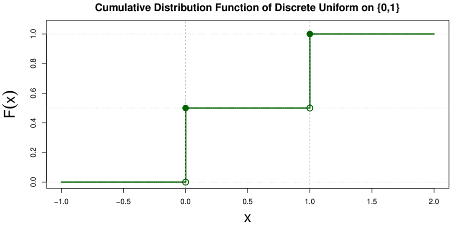
A coin flip - a discrete uniform distribution

Consider flipping a fair coin and let X denote the result of the flip (e.g., 0 for tails and 1 for heads).





Aim

$$X \in \{0, 1\}$$

$$\mathbb{P}(X = 0) = \mathbb{P}(X = 1) = \frac{1}{2}$$

Construction

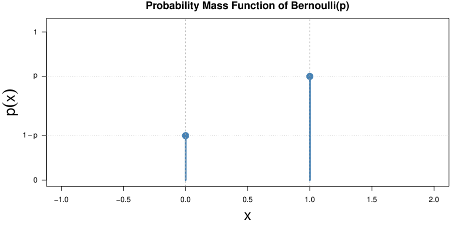
$$X = \mathbf{1}\left\{U > \frac{1}{2}\right\}$$

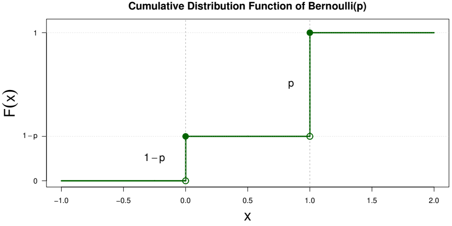
or

$$X = \lfloor 2U \rfloor$$

The general Bernoulli distribution

If we instead aim to generate a Bernoulli distributed random variable, $X \sim \text{Bernoulli}(p)$ with $\mathbb{P}(X=0) = 1-p$ and $\mathbb{P}(X=1) = p$.





Aim

$$X \in \{0, 1\}$$

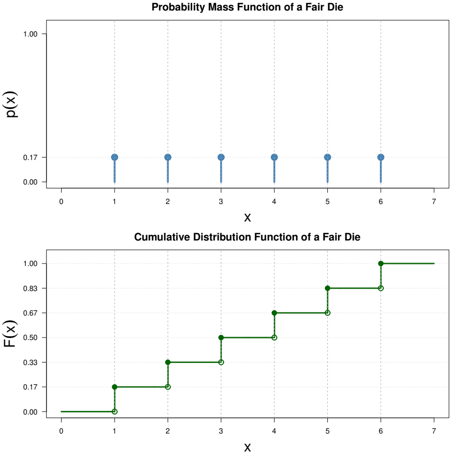
$$\mathbb{P}(X=0) = 1-p$$
$$\mathbb{P}(X=1) = p$$

Construction

$$X = \mathbf{1}\{U > 1-p\}$$

A fair die - a more general discrete uniform distribution

We now want to generate a discrete uniform distribution on the integers $\{1, \ldots, 6\}$.



Aim

$$X \in \{1,2,3,4,5,6\}$$

$$\mathbb{P}(X = i) = \frac{1}{6}, \quad i \in \{1,2,3,4,5,6\}$$

Construction

$$X = \lfloor 6U \rfloor + 1$$

This generalises to any uniform distribution on consecutive integers.

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1. Introduction
2. Simple examples
3. Methods for sampling from discrete distributions
4. Rejection methods
5. The Alias method
6. Exercise 2

Discrete distributions: Direct (crude) method

Suppose a discrete random variable, X, can take the values $x_1 < x_2 < \cdots < x_k$, with probabilities

$$p_i = \mathbb{P}(X = x_i), \qquad i = 1, \ldots, k.$$

Basic idea

Generate $U \sim U(0,1)$. The value x_i is selected whenever U falls into an interval of length p_i. More precisely,

$$X = x_i \quad \text{if} \quad \sum_{j=1}^{i-1} p_j < U \leq \sum_{j=1}^{i} p_j.$$

Equivalently,

$$X = x_i \quad \text{if} \quad F(x_{i-1}) < U \leq F(x_i),$$

where F denotes the distribution function of X.

Geometric distribution

Consider a random variable $X \sim \mathrm{Geometric}(p)$ following a geometric distribution. Then X can be interpreted as a discrete waiting (stopping) time.

Probability functions

$$f(n) = \mathbb{P}(X = n) = (1-p)^{n-1}p, \quad n \in \mathbb{N}$$
$$F(n) = \mathbb{P}(X \leq n) = 1-(1-p)^{n}, \quad n \in \mathbb{N}$$

Using the direct method

$$X = n \quad \text{if} \quad F(n-1) < U \leq F(n)$$

yields

$$X = n \quad \text{if} \quad 1-(1-p)^{n-1} < U \leq 1-(1-p)^{n}.$$

After rearranging,

$$X = n \quad \text{if} \quad n-1 < \frac{\log(U)}{\log(1-p)} \leq n.$$

This leads to the construction

$$X = \left\lfloor \frac{\log(U)}{\log(1-p)} \right\rfloor + 1.$$

Discrete distributions: Implementation

The direct (crude) method can be implemented as follows. For X taking values $x_1 < x_2 < \cdots < x_k$, with probabilities $p_i = \mathbb{P}(X = x_i)$, $i = 1, \ldots, k$:

Algorithm

1. Generate $U \sim U(0,1)$.
2. Find the interval i containing U, i.e., $P_{i-1} < U \leq P_i$, where $P_i = \sum_{j=1}^{i} p_j$.
3. Output $X = x_i$.

Implementation strategies

- Linear search
- Rearrangement of intervals
- Binary search
- Indexed search

The choice of search strategy can have a significant impact on computational efficiency when (k) is large.

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Simple rejection

The rejection method provides an alternative way of generating discrete random variables. Assume X takes values in $\{1, 2, ..., k\}$ with probabilities $P(X = i) = p_i$.

Choose a constant c such that

$$c > \max_i \{p_i\}.$$

Algorithm

1. Generate $I = \lfloor kU_1 \rfloor + 1$.
2. If $U_2 \leq p_I / c$, then $X = I$, else go to step 1.

The probability of obtaining the outcome i is

$$\frac{\frac{1}{k} \frac{p_i}{c}}{\sum_{j=1}^{k} \frac{1}{k} \frac{p_j}{c}} = p_i.$$

Hence, the generated variable has the desired distribution.

General rejection method

Suppose we wish to generate a discrete random variable X with pmf $P(X = i) = p_i$ and that we already have access to random variables Y with pmf $P(Y = i) = q_i$.

Choose a constant c such that

$$c > \max_i \{p_i/q_i\}.$$

Algorithm

1. Generate Y according to the proposal distribution $\{q_i\}$: $I = Y$.
2. If $U \leq p_I/cq_I$, then $X = I$, else go to step 1.

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Alias method

The alias method is a technique for generating random variables from a general discrete distribution.

Characteristics

- Generates random variables from an arbitrary discrete distribution.
- Transforms a discrete uniform distribution into a general discrete distribution.
- Requires only:
  - one random number generation,
  - one comparison,
  - and potentially one table lookup.
- Very efficient during simulation.

Drawback

A preprocessing step is required to construct the alias tables, making the setup procedure more complicated than for the direct or rejection methods.

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Exercise 2

In this exercise, you may use built-in procedures for generating random numbers. Compare the results obtained from simulation with the corresponding theoretical results. Use histograms and statistical tests where appropriate.

Part 1

Choose a value for the probability parameter (p) in the geometric distribution and simulate 10,000 observations.

- Compare the simulated and theoretical distributions. You should experiment with small, moderate, and large values of p.

Exercise 2 (continued)

Part 2

Consider the six-point distribution

x_i	1	2	3	4	5	6
$\mathbb{P}(X = x_i)$	$\frac{7}{48}$	$\frac{5}{48}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{4}$	$\frac{5}{16}$

Generate observations from this distribution using:

1. the direct (crude) method,
2. the rejection method,
3. the alias method.

Exercise 2 (continued)

Part 3

Compare the three generation methods using appropriate performance measures and discuss the results.

Possible criteria include:

- Computational efficiency
- Ease of implementation
- Memory requirements
- Accuracy of the generated distribution

Part 4

Provide recommendations regarding when each method should be preferred.
Discuss the advantages and disadvantages of the:

- Direct (crude) method
- Rejection method
- Alias method

If time permits, support your conclusions through additional simulation experiments.